	(38) $u_\tau R = v_\infty b$	0.3	
	(39) $v_{\infty,min} = \sqrt{\frac{2GM}{R\left(\frac{b^2}{R^2}-1\right)}}$	0.2	
	$v_{\infty,min} = 252 km/s$	0.1	
2.4	(40) $\Delta A = \frac{16}{3}\pi^2 G\rho_4^2 r^4 \Delta r$	0.3	**0.6**
	(41) $A = \frac{16}{15}\pi^2 G\rho_4^2 R^5$	0.2	
	$A = 1.33 \times 10^{45} J$	0.1	
합계			**10.0**

문제 3. 각기둥의 저항 (10.0점)

1. 수학적 준비 (3.0점)

1.1 **[0.2점]**

$$x_k = A\lambda^k. \tag{1}$$

1.2 **[0.4점]** λ^k λ^{k-1}:

$$\lambda^k = \lambda^{k-1}\cdot\lambda$$

$$\lambda^k = \left(p_k + q_k\sqrt{3}\right) = \left(p_{k-1} + q_{k-1}\sqrt{3}\right)\cdot\left(2+\sqrt{3}\right) = 2p_{k-1} + p_{k-1}\sqrt{3} + 2q_{k-1}\sqrt{3} + 3q_{k-1} = \\ = \left(2p_{k-1} + 3q_{k-1}\right) + \left(p_{k-1} + 2q_{k-1}\right)\sqrt{3}. \tag{2}$$

$$\begin{aligned} p_k &= 2p_{k-1} + 3q_{k-1} \\ q_k &= p_{k-1} + 2q_{k-1}. \end{aligned} \tag{3}$$

$$\lambda^{k-1} = p_{k-1} + q_{k-1} = \lambda^k\cdot\lambda^{-1} = \left(p_k + q_k\sqrt{3}\right)\cdot\left(2-\sqrt{3}\right) = \\ = \left(2p_k - 3q_k\right) + \left(2q_k - p_k\right)\sqrt{3}, \tag{4}$$

$$\begin{aligned} p_{k-1} &= 2p_k - 3q_k, \\ q_{k-1} &= 2q_k - p_k. \end{aligned} \tag{5}$$

1.3 **[0.7점]**

$p_0 = 1, \quad q_0 = 0$. 1 .

1.

k	p_k	q_k
0	1	0
1	2	1
2	7	4
3	26	15
4	97	56
5	362	209

1.4 **[0.2점]**

$$\lambda^{-1} = \frac{1}{2+\sqrt{3}} = 2-\sqrt{3}, \tag{6}$$